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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Electronics and Communication Engineering

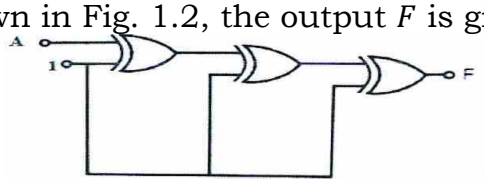
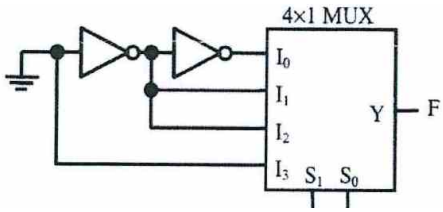
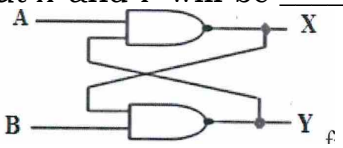
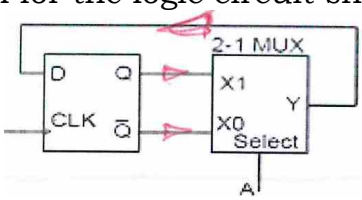
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is meant by redundant group?	01
	1.2	For the circuit shown in Fig. 1.2, the output F is given by _____.  Fig. 1.2	01
	1.3	The logic function implemented by the circuit shown in Fig. 1.3 is _____.  Fig. 1.3	02
	1.4	In the Fig. 1.4 shown with input $A = 1$, The input B is a sequence 101010…. Then the output X and Y will be _____.  Fig. 1.4	02
	1.5	The present output Q_n of an edge triggered JK flip-flop is logic '0'. If $J = 1$, then Q_{n+1} is _____.	01
	1.6	The output frequency of a decade counter that is clocked from a $50KHz$ signal is _____.	02
	1.7	Draw the state diagram for the logic circuit shown in Fig. 1.7  Fig. 1.7	02

1.8 Reduce the number of states in the following state table, and tabulate the reduced state table:

Present State	*Next State*		*Output*	
	$X=0$	$X=1$	$X=0$	$X=1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

02

1.9 Assuming that all flip-flops are in rest condition initially, the count sequence observed Q_A after 4 clock cycle in the circuit shown in Fig 1.9 is _____.

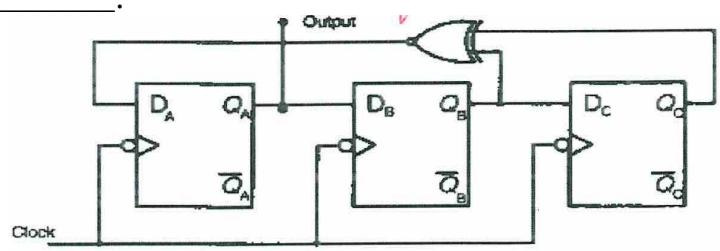


Fig. 1.9 02

1.10 An 8-bit ring counter will go through a sequence of _____ states. 01

1.11 A serial adder uses two $4-bit$ shift registers. Register A holds a binary number 1001 and register B holds 0101. The carry flip-flop Q is initially reset. List the binary values in register A and the output of flip-flop Q after each shift. Assume that sum is stored in register A. 02

1.12 The address bus width of a scratchpad memory of size 2048×16 is _____. 01

1.13 In status bit, Z is _____ is the output of ALU is _____. 01

PART-B

2 a Simplify the Boolean expression, F, and show its implementation using only $NAND$ gates. $F(A,B,C,D,E) = \sum m\,(0,1,4,5,16,17,21,25,29)$ 05

b Describe the operation of $ECL\ OR$ gate with diagram. 06

c Define fan-out and fan-in for a digital gate/circuit.
Refer to the $74ALS20\ NAND$ gate specification given below and determine how many $74ALS20\ NAND$ gates can be driven by the output of another $74ALS20\ NAND$ gates.

$I_{OH(\max)}$	$I_{IH(\max)}$	$I_{OL(\max)}$	$I_{IL(\max)}$
$0.4mA$	$20uA$	$8\ mA$	$0.1\ mA$

05

3 a Design and implement a $4-$bit carry-look-ahead adder. 08

b Implement the following Boolean function using an $8:1$ multiplexer.
$F(A,B,C,D) = A\overline{B} + BD + \overline{B}C\overline{D}$ 08

OR

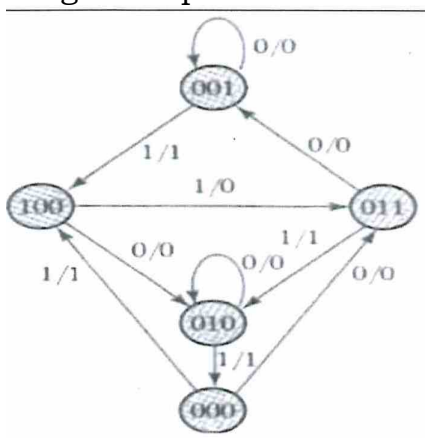
4	a	Design a 4 –bit BCD adder and explain the working.	08
	b	Construct a 4 –to–16 line decoder using five 2 –to–4 line decoder with enable inputs.	08
5	a	A sequential circuit has one input and one output. The state diagram is shown in Fig. 5.a. Design a sequential circuit using D flip-flop.  Fig. 5.a	08
	b	Design a serial adder using a sequential logic procedure (JK FFs). Draw the state table and the state diagram for the same.	08
		OR	
6	a	A PN flip-flop has four operations: clear to 0, no change, complement and set to 1, when inputs P and N are $00, 01, 10$, and 11 respectively. i) Tabulate the characteristic table. ii) Derive the characteristic equation. iii) Tabulate the excitation table. iv) Show how the PN flip-flop can be converted to a D flip-flop.	10
	b	Design a sequence generator to generate the sequence (0111) using JK flip-flops.	06
7	a	Design mod-32 ripple counter. If the propagation delay of each flip-flop is $20\,nsec$, determine the maximum clock frequency to be used.	05
	b	Design a $4-bit$ shift register with a parallel load using D flip-flops. There are two control units; shift and load. When shift= 1 the contents of the register are shifted by one position right. New data is transferred into the register when load= 1 and shift= 0. If both control inputs are equal to 0 the contents of the register do not change. Write the functional table showing different operations.	05
	c	Design a 4 –bit BCD synchronous counter using JK flip flops and draw its timing diagrams.	06
8	a	Design an arithmetic circuit with two selection variables s_1 and s_0, that performs the arithmetic operation shown in Table 8.a.1. Modify the arithmetic circuit into an ALU with mode select variable s_2. When $s_2 = 0$, the ALU is identical to the arithmetic circuit, when $s_2 = 1$ generates the logic functions according the Table 8.a.2. Draw the logic diagram of two typical stages.	

Table 8.a.1 : Arithmetic operations

s_2	s_1	s_0	$C_{in} = 0$	$C_{in=1}$
0	0	0	$F = A$	$F = A + 1$
0	0	1	$F = A + B$	$F = A + B + 1$
0	1	0	$F = A - B - 1$	$F = A - B$
0	1	1	$F = A - 1$	$F = A$

Table8.a.2: Logical operations

s_2	s_1	s_0	$C_{in} = 0$
1	0	0	$F = A \vee B$
1	0	1	$F = A \oplus B$
1	1	0	$F = A \wedge B$
1	1	1	$F = A$

b Draw the logic diagram of a $4 - bit$ combinational logic shifter. 12 04